

Week 35, Tuesday Session

Flipped classroom: your questions, then Ridge and Lasso in practice

Bring your laptop – we work in `week35tuesday.ipynb`

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How the Tuesday sessions work (weeks 35–48)

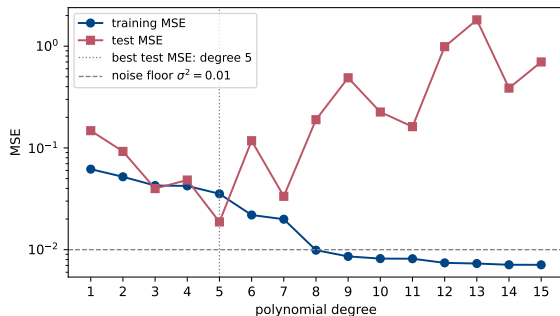
The format

Mondays are traditional lectures, with pauses for questions. Tuesdays are *flipped*: you have heard the lecture and skimmed the reading – the session belongs to your questions and to working on concrete cases.

1. **First ~20 min:** open questions from Monday's lecture, then the warm-up questions on the next slides, discussed with your neighbours and in plenum.
2. **Rest of the double session:** case work in pairs or small groups, from the companion notebook (`week35tuesday.ipynb`) and this week's exercise set. We circulate and help; short plenary wrap-ups between cases.
3. **Last 5 min:** what to finish before the exercise deadline, and what Monday next week builds on.

The cases are chosen so that the code you write here is directly reusable in the weekly exercises and in Project 1.

Picking up from Monday: the exercise-3 figure



Monday's lecture showed this figure – training and test MSE for polynomial fits of degree 1 to 15 (exercise 3 of this week's set). Before the warm-ups, discuss with your neighbour:

- ▶ Where in this figure is *underfitting*, where *overfitting*, and what defines the boundary?
- ▶ The training MSE dips below the noise floor $\sigma^2 = 0.01$. What is the model "learning" there?
- ▶ Your run of the notebook uses a different random split. Which features of the figure survive a new split, and which move around?

You will reproduce and extend this figure in Case 4 – with Ridge's λ replacing the degree as the complexity dial.

Warm-up 1: the one-sentence summaries

Question

Complete each sentence from Monday, in one line each, without notes: (a) Gauss–Markov says OLS is optimal among ...; (b) Ridge beats OLS in mean squared error because ...; (c) the Lasso, unlike Ridge, can ...; (d) before applying any penalty, the features must be ...

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Answer

(a) ...estimators that are *linear in y and unbiased* – biased estimators were never in the competition. (b) ...a little bias costs $O(\lambda^2)$ while the variance falls linearly in λ – so the MSE slope at $\lambda = 0$ is strictly negative (Hoerl–Kennard). (c) ...set coefficients *exactly* to zero at finite λ – selection, not just shrinkage. (d) ...standardised on the *training data* (and the intercept left unpenalised).

Warm-up 2: read the formula

Question

The Ridge fit shrinks the coordinate of y along u_i by $\sigma_i^2/(\sigma_i^2 + \lambda)$. For a badly conditioned degree-10 Vandermonde matrix, which directions are affected first as λ grows from zero, and why is that *exactly* what we want?

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Answer

The trailing directions, $\sigma_i^2 \ll \lambda$: they are the combinations of columns the data barely determine, and their OLS coefficients carry variance σ^2/σ_i^2 – the largest by far. Ridge suppresses first the directions where the fit is mostly noise, and touches the well-determined leading directions last. The penalty is a variance filter ordered by the singular spectrum.

Warm-up 3: predict before you compute

Question

In Case 2 you will fit the degree-5 polynomial with Ridge for $\lambda \in \{0, 10^{-4}, 10^{-2}, 1, 100\}$. Before running anything: what do you expect to happen to (a) the size of the coefficient vector $\|\hat{\theta}\|_2$, (b) the training MSE, (c) the test MSE, as λ increases? Sketch the three curves.

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Answer

(a) Strictly decreasing (the norm of the Ridge solution is monotone in λ). (b) Strictly increasing: OLS is by construction the training-MSE minimiser, every $\lambda > 0$ can only do worse *on training data*. (c) The interesting one: typically falls first (variance reduction), passes through a minimum, then rises (bias takes over) – the U-shape of the bias–variance trade-off. If your test-MSE curve does not turn up again, extend the λ grid. Verify all three this session.

Case 1: Section 3.13 live – scaling and the intercept in OLS, Ridge, Lasso

A second-order polynomial, simple enough that every coefficient is interpretable:

```
n = 100
x = rng.random(n) # x in [0, 1]
y = 2.0 + 5.0*x**2 + 0.1*rng.standard_normal(n)
X = np.column_stack([np.ones(n), x, x**2]) # intercept, x, x^2
```

1. **OLS**: fit with the intercept column; then fit on *centred* data without intercept and recover $\hat{\theta}_0 = \bar{y} - \sum_j \bar{x}_j \hat{\theta}_j$. Same predictions? Now rescale $x \rightarrow 1000x$ (“millimetres”): what happens to the coefficients, and what to the predictions?
2. **Ridge** ($\lambda = 0.1$): (a) the naive fit – raw columns, intercept column included *and penalised*; (b) the proper fit – standardised columns, centred y , intercept recovered, not penalised. Compare $\hat{\theta}_0$ and the predictions. Then shift all targets, $y \rightarrow y + 100$: which of the two fits shifts with the data?
3. **Lasso** (same λ ; mind the `alpha` conversion from Case 2): repeat (a) and (b). On the *unscaled* millimetre columns, which coefficients does the Lasso zero out – and does the model still make sense?

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Hint: Expected findings: OLS predictions are invariant under both tricks – the intercept is just another column. A penalised intercept refuses to follow $y \rightarrow y + 100$ (the fit is dragged towards zero); unscaled columns make the penalty act with units, and the Lasso may kill the “wrong” feature entirely. The intercept sets the *level* of the data – it must never pay a penalty.

Case 2: own Ridge vs. scikit-learn – match the conventions

Write your own Ridge solver and reconcile it with the library, to machine precision.

```
def ridge_svd(X, y, lambda):  
    U, s, Vt = np.linalg.svd(X, full_matrices=False)  
    return Vt.T @ (s / (s**2 + lambda) * (U.T @ y)) # sum_i s_i/(s_i^2+lambda) (u_i.y) v_i
```

1. Verify `ridge_svd` against $(X^T X + \lambda I)^{-1} X^T y$ on centred data.
2. Now match `sklearn.linear_model.Ridge(alpha=?)`: which `alpha` reproduces your λ ? Check with `np.allclose`.
3. Repeat for Lasso: which `alpha` corresponds to our λ convention? (It is *not* the same rule as for Ridge.)
4. Increase the polynomial degree to 12 and compare `ridge_svd` with the normal-equations version at $\lambda = 0$ and $\lambda = 10^{-8}$. What does the tiny penalty do to the conditioning?

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Hint: Monday's "convention trap" insight has the dictionary. For (4): $\kappa(X^T X + \lambda I) \leq (\sigma_{\max}^2 + \lambda)/\lambda$.

Case 3: shrinkage vs. selection on real columns

Reproduce the coefficient-path figure from Monday, then make your own on a correlated design.

1. Verify the toy problem analytically *and* numerically: $y = (4, 2, 3)^T$, X diagonal with entries 2, 1; Ridge paths $8/(4 + \lambda)$, $2/(1 + \lambda)$; Lasso zeros at $\lambda = 4$ and 16.
2. Build a design whose columns x^4 and x^5 (on $[0, 1]$) are strongly correlated. Plot Ridge and Lasso paths over $\lambda \in [10^{-4}, 10^2]$. Which coefficients does the Lasso kill first? Does Ridge ever kill any?
3. For one λ , check the Lasso optimality condition: for every zeroed coefficient, is $|\frac{2}{n}x_j^T r| \leq \lambda$?

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Hint: Use `np.corrcoef` to see how correlated the columns are. The KKT check in (3) is two lines and is the best unit test a Lasso solver has.

Case 4: model selection: the U-curve with and without a penalty

Reproduce Monday's exercise-3 figure, then extend it (Hastie et al., Fig. 2.11):

```
x = np.linspace(-3, 3, n).reshape(-1, 1)
y = np.exp(-x**2) + 1.5*np.exp(-(x - 2)**2) + rng.normal(0, 0.1, x.shape)
```

1. OLS: plot training and test MSE against polynomial degree $d = 1, \dots, 15$. Locate the optimal degree.
2. Fix $d = 15$ and switch to Ridge: plot training and test MSE against λ on a log grid. Compare the best test MSE with the best OLS degree from (1).
3. Which knob – degree or λ – gives the smoother path from underfitting to overfitting? Relate to $\text{df}(\lambda) = \sum_i \sigma_i^2 / (\sigma_i^2 + \lambda)$: compute df for your best λ and compare with the best degree.

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Hint: Two complexity dials, one phenomenon. $df(\lambda)$ makes them comparable on the same axis – a preview of next week's bias-variance analysis.

Wrap-up, deadlines, and Monday

- ▶ **This week's exercise set** (deadline Friday at midnight): the analytical matrix–vector derivatives, your own OLS code with `scikit-learn` comparison, and the train/test split – Cases 2 and 4 give you most of the code.
- ▶ **Keep your functions:** `ridge_svd`, the scaling helper and the path plots go straight into Project 1. And keep Case 1's conclusion in mind for every penalised fit you ever run: standardise the features, never penalise the intercept.
- ▶ **If you finished everything:** implement the coordinate-descent Lasso from Monday's slides and check it against `sklearn` on the correlated design of Case 3.
- ▶ **Monday next week:** resampling methods – the bootstrap and cross-validation – and the bias–variance trade-off derived in full. The U-curves you produced today are the experimental data for that lecture.

Questions in the meantime

Use the discussion forum, or bring them to next Tuesday – the first twenty minutes are always yours.